PROJECT-BASED LEARNING

PHASE-I EVALUATION

Fund Nexus: A Unified Platform for Transparent Fund Management & Donation Tracking

Team ID: DSCPP-III-2026-T101

Department of Computer Science & Engineering
Graphic Era (Deemed to be University), Dehradun
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Team & Mentor Details

Team Information

Team ID: DSCPP-III-2026-T101
Semester: 3rd
Domain: FinTech/Fund Management System

Team Members

1. *Yashasvi Sharma – 2510010351 (Team Lead)*
2. *Pari Singhal – 2510012558*
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Mentor

Mr. Yuvraj Joshi

Interactions completed: _____

Problem Statement & Motivation

Problem Statement

Donors and welfare organizations (NGOs, community funds, crowdfunding campaigns) lack a single, transparent system to raise, track, and audit funds. Contributions are scattered across cash, bank transfers, and multiple apps, making it hard to verify how much was raised, how it was spent, and whether the campaign is genuine.

Motivation

- No unified ledger of donations vs. disbursements
- Low donor trust due to lack of transparency
- Manual record-keeping is error-prone

Target Users / Stakeholders

- Donors contributing to campaigns/funds
- Fund/campaign administrators (NGOs, clubs, causes)
- Auditors / platform admin reviewing

Objectives

Project Objectives

1. Design a centralized system where multiple funds/campaigns can be created and tracked.
2. Implement donor registration, contribution recording, and expenditure logging.
3. Apply data structures (linked lists, trees, hashing) to manage funds, donors, and transaction history efficiently.
4. Use OOP principles to model Fund, Donor, Transaction, and Campaign as reusable, extensible classes.
5. Generate real-time balance summaries and transparent transaction reports per fund.

Key Objective: Build a working prototype that lets any user create

Proposed Solution

Proposed Approach

1. Donor / Admin registers and logs in
2. Create or select a Fund (campaign)
3. Record a Donation or Expense transaction
4. Update fund ledger and running balance
5. Display transaction history & summary report

Key Features

- Multiple funds/campaigns under one platform
- Donor contribution history & receipts
- Real-time balance and expenditure tracking
- Searchable, sortable transaction ledger
- Exportable summary report per fund

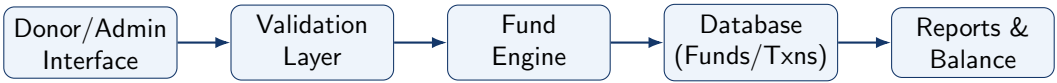
Fund Nexus acts as a single "nexus" connecting donors, funds, and expenditures through one transparent ledger.

Integration of PBL Course Concepts

Course	Concepts Applied	Project Module
Data Structures in C	Linked lists for transaction history, BST/hashing for fast donor & fund lookup, sorting for report generation	Transaction Ledger, Donor Registry
OOPs with C++	Classes for Fund, Donor, Transaction, Campaign; inheritance for fund types; STL containers (vector, map) for in-memory storage	Core Fund Management Engine
Operating Systems	Synchronization / mutex-style locking to prevent race conditions on concurrent donations to the same fund	Concurrent Transaction Handling
DBMS	Relational schema (Donors, Funds, Transactions), SQL queries, transactions & normalization for persistent, consistent storage	Database Layer

Requirement: Demonstrate meaningful integration of both subjects in **one** project.

System Architecture / Workflow



Major Components

- User/Admin Interface (console or web UI)
- Fund Management Engine (core logic, classes)
- Persistent Database Layer (donors, funds, transactions)

Data / Control Flow

- Donor submits contribution/expense request
- Fund Engine validates and updates ledger
- Database persists transaction; report module reads and displays updated balance

Technology Stack & Methodology

Technology Stack

- Programming: C++ (Core Logic, OOP)
- Database: MySQL
- Tools / Framework: STL, (optional) simple CLI/GUI
- IDE: VS Code / Code::Blocks
- Version Control: Git / GitHub

Development Methodology

1. Requirement Analysis
2. System Design
3. Implementation
4. Testing
5. Evaluation

C++ was chosen for strong OOP support and STL data structures; MySQL provides reliable, queryable persistence for auditable transaction records.

Initial Progress & Team Contribution

Progress Achieved

- Studied existing crowdfunding/ledger platforms
- Finalized requirements & feature list
- Designed system architecture and DB schema
- Implemented initial Fund and Transaction classes
- Set up sample donor/fund test data

Individual Contribution

Member	Role	Contribution
M1	Team Lead	Architecture design, Fund Engine core logic
M2	Developer	Donor & Transaction classes, STL integration
M3	DB / Testing	Database schema, SQL queries, test cases
M4	Developer / Docs	UI/CLI, reports, documentation

Roadmap, Expected Outcomes & References

Roadmap

1. Phase-I: Proposal & Design
2. Phase-II: Development
3. Phase-III: Final Implementation

Expected Outcomes

- Working Fund Nexus prototype
- Accurate, real-time fund balances
- Transparent, auditable transaction history
- Scalable to multiple funds/organizations

References

1. Data Structures & OOP course textbooks
2. MySQL / C++ STL official documentation
3. Research papers on transparent crowdfunding systems
4. GitHub repositories on donation-tracking apps

Thank You

Questions & Suggestions